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(72) Inventors; and

(71) Applicants: **AGHAZADEH, Hamed** [IR/IR]; No. 15, 3rd floor, Dr. Hesabi St., Shabnam Square, West Sahand Square, Rajaee shahr, 5166835363 (IR). **KAYA, Ali** [TR/TR]; Barbaros hayrettin pasha area, 1988 st, NO 4, papatya1 residence, floor 3, unit 36, Esenyurt, 34522 Istanbul (TR). **SHAMLOUEI, Mostafa** [IR/IR]; No. 20, corner of Alipour Alley, North Jannatabad, Tehran, 1478778771 (IR). **ABOULHASSANZADEH, Sobhan** [IR/IR]; 20 Meters Bahman St, Khoy, 5816655611 (IR). **ABOULHASSANZADEH, Samin** [IR/IR]; 257 Moorefield Park Newbridge Co.Kildare, Kildare, W12HE27 (IR). **HOMAYOON, Shidoosh** [IR/IR]; No. 20, 3rd floor, unit 6, Mehr Dd en, 3rdMahdi St, Southern Eta'ati St, Marzdaran blvd, Tehran, 1464778448 (IR). **ZAREI, Alireza** [IR/IR]; No 40, Shahid Khosrozadeh Street, Laleh Avenue, Shiraz, 7168657847 (IR). **KARIMI, Parvin** [IR/IR]; Golestan Town, Dehkhoda Blvd., Corner of Deh Alley, Shiraz, 7189916485 (IR). **PARSA, Mahsa** [IR/IR]; Third floor, block 18, Nastaran complex, sixth street, Manzariye, Tabriz, 5174964517 (IR). **GHADERMARZI, Parastoo** [IR/IR]; Kermanshah, Iran, Kermanshah, 1641678451 (IR). **SADEGHI, Fatemeh** [IR/IR]; Taha building, lane 4,street 10, Southern Iman Avenue, Shiraz, 7188773579 (IR). **SALMANIOMOUMI, Reza** [IR/IR]; ALBORZ, IRAN, Alborz, 4114245781 (IR).

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(54) **Title:** HOCI-PFP-ALGINATE-CHITOSAN NANOPARTICLES PRODUCTION METHOD AS A LUNG CANCER ANTIBODY

(57) **Abstract:** This invention is related to the fields of medical biotechnology, nanobiotechnology, and nanomedicine. It involves a nanomedicine product that can release HOC1 (hypochlorous acid) at the tumor site in a controlled manner to treat lung cancer. HOC1 is a potent antimicrobial and antitumor agent that can target tumor tissues while minimizing damage to healthy tissues. The produced nanoparticles have a two-layer structure: the inner layer contains alginate, PFP (perfluoropentane), and $Ca(OCl)_2$ to release HOC1, and the outer layer has a chitosan coating with folate and salicylic acid for targeting and proton release. The production process involves nanoemulsion and coating with the ionic gelation technique. When the nanoparticles are delivered to the lungs and targeted to cancer cells with folate, ultrasound radiation triggers the release of HOC1 to induce apoptosis and destroy cancer cells effectively, reducing side effects and improving treatment outcomes for lung cancer patients.

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Title of Invention: HOCI-PFP-Alginate-Chitosan Nanoparticles Production Method as a Lung Cancer Antibody

[0001] Lung cancer is one of the most common and deadly types of cancer worldwide, and due to its specific complications, its treatment is still challenging. In 2022, lung cancer will be responsible for the death of about 2.09 million people worldwide. The main types of lung cancer include non-small cell carcinoma (NSCLC) and small cell carcinoma (SCLC). Because lung cancer is often diagnosed at an advanced stage, existing treatments for this disease typically face problems including severe side effects and limited effectiveness. HOCl nanomedicine product with controlled release capability at the tumor site for lung cancer treatment is especially used in medical and pharmaceutical industries. As one of the most common types of cancer, lung cancer accounts for approximately 14% of new cancer cases worldwide. Considering the importance of hypochlorous acid (HOCl) in cancer treatment and the potential benefits of its delivery by inhalation, the development of a new drug delivery system that can provide HOCl in a sustained and controlled release manner and minimize side effects is of utmost importance. It is very important. This nanoparticle has a two-layer structure: the inner layer: contains alginate along with perfluoropentane (PFP), which acts as a phase change agent, and calcium hypochlorite is located in the alginate core of the nanoparticle as a source of hypochlorite anion. The outer layer includes chitosan coating conjugated with folate and salicylic acid. The folate in the chitosan coating helps to effectively target the nanoparticle to the cancer cells, and the salicylic acid acts as a proton source.

[0002] The production process of this nanoparticle uses a two-step method including nanoemulsion and coating using ionic gelation technique. After inhaled nanoparticle delivery to the lung, the nanoparticles are specifically targeted to cancer cells using the properties of folate on the surface of chitosan. After targeting, the radiation of ultrasound waves changes the phase of perfluoropentane and increases the internal pressure of the nanoparticle, which leads to the collapse of the nanoparticle structure and the release of HOCl. This HOCl is released with a specific concentration to the tumor site induces apoptosis and eventually destroys the cancer cells. By using this advanced drug delivery system, it is possible to achieve more effective lung cancer treatment by reducing side effects and improving the efficiency of HOCl delivery to the tumor.

Technical Field

[0003] A61K 39/00 - C12Q 1/6886 - A61K33/00 - G01N 33/537

Background Art

[0004] 1- Hypochlorous acid selectively promotes toxicity and the expression of danger signals in human abdominal cancer cells

[0005] 22 March 2021

[0006] Tumors of the abdominal cavity, such as colorectal, pancreatic and ovarian cancer, frequently metastasize into the peritoneum. Large numbers of metastatic nodules hinder curative surgical resection, necessitating lavage with hyperthermic intraperitoneal chemotherapy (HIPEC). However, HIPEC not only causes severe side effects but also has limited therapeutic efficacy in various instances. At the same time, the age of immunotherapies such as biological agents, checkpoint- inhibitors or immune-cell therapies, increasingly emphasizes the critical role of anticancer immunity in targeting malignancies. The present study investigated the ability of three types of long-lived reactive species (oxidants) to inactivate cancer cells and potentially complement current HIPEC regimens, as well as to increase tumor cell expression of danger signals that stimulate innate immunity. The human abdominal cancer cell lines HT-29, Panc-01 and SK-OV-3 were exposed to different concentrations of hydrogen peroxide (H2O2), hypochlorous acid (HOCl) and peroxynitrite (ONOO-). Metabolic activity was measured, as well as determination of cell death and danger signal expression levels via flow cytometry and detection of intracellular oxidation via high-content microscopy. Oxidation of tumor decreased intracellular levels of the antioxidant glutathione and induced oxidation in mitochondria, accompanied by a decrease in metabolic activity and an increase in regulated cell death. At similar concentrations, HOCl showed the most potent effects. Non-malignant HaCaT keratinocytes were less affected, suggesting the approach to be selective to some extent. Pro-immunogenic danger molecules were investigated by assessing the expression levels of calreticulin (CRT), and heat-shock protein (HSP)70 and HSP90. CRT expression was greatest following HOCl and ONOO- treatment, whereas HOCl and H2O2 resulted in the greatest increase in HSP70 and HSP90 expression levels. These results suggested that HOCl may be a promising agent to complement current HIPEC regimens targeting peritoneal carcinomatosis.

[0007] The mentioned research was published in the journal Oncology Reports and showed that HOCl can selectively increase toxicity and expression of danger signals in cancer cells. In this study, HOCl was compared with hydrogen peroxide (H2O2) and peroxynitrite (-ONOO). The results showed that HOCl is the most potent reactive oxygen species (ROS) in reducing metabolic activity and increasing cell death, and could be used as a promising adjunct to intraperitoneal chemotherapy (HIPEC) regimens and enhancing antitumor immune responses, we used the HOCI in our

method to create an antibody exclusively for lung cancer and we processed a different method under the influence of ultrasound waves in it.

[0008] 2- Inhibiting lysine 353 oxidation of GRP78 by a hypochlorous probe targeting endoplasmic reticulum promotes autophagy in cancer cells

[0009] 12 November 2019

[0010] The level of hypochlorous acid (HOCl) in cancer cells is higher than that in non-cancer cells. HOCl is an essential signal for the regulation of cell fate and works mainly through the protein post-translational modifications in cancer cells. However, the mechanism of HOCl regulating autophagy has not been clarified. Here we reported that a HOCl probe named ZBM-H targeted endoplasmic reticulum and induced an intact autophagy flux in lung cancer cells. Furthermore, ZBM-H promoted the binding of GRP78 to AMPK and increased the phosphorylation of AMPK in a dose- and time-dependent manner. GRP78 knockdown inhibited ZBM-H-induced AMPK phosphorylation and ZBM-H-stimulated autophagy. In addition, mass spectrometry combined with point mutation experiments revealed that ZBM-H increased GRP78 activity by inhibiting HOCl-induced lysine 353 oxidation of GRP78. Following ZBM-H treatment in vitro and in vivo, cell growth was significantly inhibited while apoptosis was induced. Nevertheless, exogenous HOCl partially reversed ZBM-H-inhibited cell growth and ZBM-H-induced GRP78 activation. In brief, we found that an endoplasmic reticulum-targeted HOCl probe named ZBM-H, acting through attenuating HOCl-induced GRP78 oxidation, inhibited tumor cell survival by promoting autophagy and apoptosis. Overall, these data demonstrated a novel mechanism of hypochlorous acid regulating autophagy by promoting the oxidation modification of GRP78.

[0011] The mentioned study was published in Cell Death & Disease and identified HOCl as an autophagy inducer in cancer cells. The research showed that HOCl increases autophagy by blocking the oxidation of lysine 353 of GRP78 in the endoplasmic reticulum and was introduced as a new therapeutic target for cancer, which is the same as our claimed method, but they comprise different path because our claimed antibody aims for lung cancer.

[0012] 3- N-Chloroamino acids mediate the action of hypochlorite on A549 lung cancer cells in culture

[0013] 11 April 2010

[0014] Hypochlorous acid, a chlorinating and oxidative agent, has been reported to be implicated in many pathologies. Its markers were found under inflammatory conditions and, at least some of it reveals biological activity. Thus, in this paper we examined whether N-chloroamino acids may act as mediators of the action of hypochlorous acid in cell culture. N-Chloroamino acids were found to possess lower oxidative

capacity than HOCl/OCl– just after addition to the growth medium. However, all the chlorocompounds studied were cytotoxic to A549 cells, induced a dose-dependent increase in the G0/G1 fraction with simultaneous reduction in the G2/M fraction, collapse of the mitochondrial potential and caspase-dependent apoptosis. The content of cellular thiols decreased after 1-h incubation with the chlorocompounds studied. Although amino acids act as scavengers of hypochlorite in plasma, the chlorinated products formed stay reactive and the pattern of their action on cells in vitro is similar to that of hypochlorite.

[0015] This paper was published in Toxicology and investigated the inhibitory effects of HOCl on the growth of A549 lung cancer cells. The results showed that HOCl has negative effects on lung cancer cells through the induction of N-chloramine acids-mediated apoptosis and investigated the underlying mechanisms of HOCl in this regard; this result helps our claimed method because we released HOCI and made an antibody for lung cancer with it but overall, these two studies are different in experiments.

[0016] 4- WO2017180815A1

[0017] Treatment of cancer with hypochlorous acid

[0018] The present invention relates to hypochlorous acid compositions and their therapy for cancer patients.

[0019] The mentioned patent explores the use of HOCl in cancer treatment. It suggests that HOCl could be combined with other cancer treatments to increase the effectiveness of the treatments and reduce side effects. HOCl could also help prevent or improve symptoms associated with cancer and its treatment. The mentioned patents are really similar to our claimed one, but their methods vary because we used ultrasound radiations and other experiments to create the claimed antibody.

Summary of Invention

[0020] A nanomedicine is developed using HOCl-PFP-Alginate-Chitosan nanoparticles to deliver HOCl to the tumor site, causing apoptosis in A549 cells and reducing lung cancer tumor growth with ultrasound irradiation. The nanoparticles are created through a two-step process involving nanoemulsion and coating. Folate-conjugated chitosan is used in the formulation. The nanoparticle size, zeta potential, encapsulation efficiency, and release profile of HOCl are studied. Cytotoxicity experiments show that the nanoparticles are effective in reducing cell viability, especially when combined with ultrasound. Protein expression analysis indicates an increase in apoptotic proteins and a decrease in anti-apoptotic proteins in A549 cells treated with the nanoparticles. In an orthotopic mouse model of lung cancer, the nanoparticles lead to tumor regression, increased survival rate, and increased apoptosis in tumors. The distribution and

accumulation of nanoparticles in various organs are examined, with significant accumulation in the lungs. Overall, the HOCl-PFP-Alginate-Chitosan nanoparticles show promise as a targeted therapy for lung cancer, especially when combined with ultrasound irradiation.

Technical Problem

[0021] A. The main problems in the treatment of lung cancer are:

[0022] 1. Late diagnosis: Approximately 50% of patients have metastatic disease at the time of diagnosis, leading to a sharp drop in survival rates to 10% at five years.

[0023] 2. Metastasis: Cancer cells can migrate to distant tissues, and brain, bone, and other organ metastases complicate management and treatment.

[0024] 3. Limitations of current treatments: Chemotherapy and radiotherapy, which are often used as primary treatments, are associated with problems such as severe side effects and reduced patient quality of life.

[0025] 4. Inadequate drug delivery: Intravenous administration of drugs for lung cancer leads to unstable drug distribution in the tumor and increased systemic side effects.

[0026] B. Statement of the objectives of the invention:

[0027] This invention deals with the design and development of a nanomedicine product with the ability to control the release of HOCl in the tumor site for the treatment of lung cancer. The main objectives of this invention are as follows:

[0028] 1. Controlled release: HOCl design of a nanomedicine system that can release HOCl in a controlled and targeted manner at the tumor site. This feature helps to improve the effectiveness of treatment and reduce side effects.

[0029] 2. Increasing the precision of treatment: by using nanomedicines, HOCl is directly transferred to the tumor, and through pulmonary delivery, especially inhalation, it increases the accumulation and retention of the drug at the tumor site. This method reduces systemic distribution and unwanted side effects.

[0030] 3. Improving anti-tumor effects: HOCl helps to destroy cancer cells and induces apoptosis in cancer cells due to its strong oxidative properties and ability to induce oxidative stress. This helps to improve treatment response and reduce tumor growth.

[0031] 4. Reduction of drug resistance: by using a nanomedicine system that can control the release of HOCl, it is possible to effectively use the optimal therapeutic concentrations at the tumor site and thus reduce the risk of drug resistance.

[0032] 5. Reduction of side effects: By focusing the drug on the tumor site and reducing the distribution in healthy tissues, systemic side effects are reduced. This helps to improve the quality of life of patients and reduce the negative side effects caused by high doses.

[0033] 6. Design and manufacture of new nanoformulations: development of nanoparticles and drug delivery systems that can effectively transfer HOCl to the tumor and have high accuracy and control.

[0034] By providing an advanced nanomedicine system with controlled HOCl release, this invention hopes to significantly improve lung cancer treatment results and the quality of life of patients with this disease.

[0035] Currently, existing treatments for lung cancer are facing serious problems and limitations. One of the major challenges in this field is the instability and insufficient efficiency of anticancer agents such as hypochlorous acid (HOCl) in clinical treatments. HOCl is known as an anticancer agent with a high capacity to induce oxidative stress and apoptosis in cancer cells, but its instability and problems related to its effective delivery to tumor tissue limit its clinical application.

[0036] c. The main problems in using HOCl to treat lung cancer are:

[0037] 1. Instability of HOCl: Due to its specific chemical properties and high reactivity, HOCl is unstable and decomposes quickly, which reduces its effectiveness in clinical treatments.

[0038] 2. Difficulty in effective delivery: It is difficult to deliver HOCl locally and inappropriate concentrations in tumor tissue. Traditional intravenous injections lead to unstable drug distribution in tumor tissue and increase side effects in healthy tissues.

[0039] 3. Systemic side effects: Using high doses to reach effective therapeutic concentrations can lead to severe side effects and reduce the quality of life of patients.

[0040] 4. Uncontrolled release: The impossibility of accurate and continuous control of HOCl release at the tumor site leads to the failure to fully exploit the antitumor capacity of this substance.

[0041] In general, it can be said that this invention deals with the design and development of a nanomedicine product with the ability to control the release of HOCl for the treatment of lung cancer. Therefore, the main objectives of this invention are as follows:

[0042] 1. Design of suitable nanomedicine for pulmonary delivery:

[0043] Design of alginate nanoparticles containing perfluoropentane (PFP) and chitosan coating conjugated with folic acid as a nano drug system for targeted delivery of HOCl to the lungs. This design increases the accuracy and effectiveness of HOCl delivery to the tumor site and reduces its non-target distribution to healthy tissues.

[0044] 2. Stable production of HOCl at the site of cancer:

[0045] Development of nanoparticles capable of encapsulating and maintaining the stable structure of HOCl in tumor tissue. This design allows the nanoparticles to continuously release HOCl at a defined concentration at the tumor site, even under conditions where HOCl is naturally unstable.

[0046] 3. Control and regulation of drug release:

[0047] Designing a controlled release system for HOCl using advanced technologies such as blasting caused by ultrasound waves to activate nanoparticles and release HOCl at precise concentrations and at appropriate times. This mechanism helps to regulate and precisely control the release of HOCl at the tumor site.

[0048] 4. Increasing the efficiency of drug delivery:

[0049] Improving the efficiency of HOCl delivery through pulmonary delivery and using nanoparticles to increase drug accumulation in tumor tissue and reduce systemic side effects. This approach helps reduce the need for high doses and reduce associated side effects.

[0050] This invention aims to provide an advanced nanomedicine system for the treatment of lung cancer, which is capable of targeted and sustained delivery of HOCl with minimal side effects. By using this system, there is hope to improve treatment results and the quality of life of lung cancer patients.

Solution of Problem

[0051] A. Fabrication of alginate-chitosan nanoparticles:

[0052] 1. Manufacturing alginate nanoparticles containing PFP:

[0053] Mixing:

[0054] To make alginate nanoparticles containing PFP, first, 1 ml of 5% calcium chloride solution containing 0.1% polyvinyl alcohol is mixed with different concentrations of Ca(OCl)2 solution and 1-5% Perfluoro Pentane (PFP). This mixture is stirred using a sonicator probe to achieve complete mixing.

[0055] • Production of nanoparticles:

[0056] After complete mixing, this compound is added to the solution of sodium alginate with a concentration of 1.23% and trehalose with a concentration of 8.25% at pH = 5 with the help of a jet nebulizer. This process leads to the formation of alginate nanoparticles containing PFP.

[0057] 2. Making chitosan conjugated with folate:

[0058] • Preparation of folate 44: mg of folate is dissolved in 15 mL of anhydrous DMSO and then EDC (Dimethylaminopropyl)-N′-ethylcarbodiimide hydrochloride) is added to the solution. This mixture is stirred for one hour at room temperature.

[0059] • Conjugation 5: millimolar chitosan sodium acetate 5% volume by weight with pH = 5 is added to the solution and stirred for 16 hours at 30°C. This action allows the carboxylic groups of folate to react with the amine groups of chitosan and form an amide bond.

[0060] • Dialysis and drying:

[0061] After adjusting the pH to 9.0, the reaction mixture is dialyzed for 3 days in sodium phosphate buffer solution with pH = 7.4 and then stored and dried at -48°C for 12 hours.

[0062] 3. Salicylic acid loading and nanoparticle coating:

[0063] • Loading:

[0064] Chitosan conjugated with folate is dissolved in a PBS solution containing 2.5 mM salicylic acid.

[0065] • Coating:

[0066] A volume of 4 ml of chitosan solution containing salicylic acid is slowly added to the alginate solution while stirring (at 500 rpm). After this, the suspension is centrifuged at 20,000 rpm for 30 minutes and then washed three times to ensure the removal of impurities and excess reagents.

[0067] b. Assessment and analysis of nanoparticles:

[0068] 1. Measurement of conjugated folate content:

[0069] Folate content in HOCl-PFP-Alginate-Chitosan nanoparticles is determined using the fluorometric method and multi-mode microplate reader. For this purpose, 2 mg of nanoparticles are dissolved in a mixture of DMSO and dichloromethane, and then the amount of folate is measured and calculated using a calibration curve. The amount of folate conjugated to chitosan was calculated as 78.5% ± 0.09.

[0070] 2. Examining the properties of nanoparticles:

[0071] • Size and zeta potential: size distribution and zeta potential of nanoparticles are checked using the Malvern Zeta Nano ZS device. The average hydrodynamic size of the nanoparticles was measured to be 1092 ± 170.3 nm and the relatively low dispersion index (PDI) (<0.3) indicates a small size distribution of the nanoparticles.

[0072] • Encapsulation efficiency: Using high-performance liquid chromatography (HPLC), the encapsulation efficiency of HOCl-PFP-Alginate-Chitosan-folate-NPs nanoparticles has been determined to be 98.38%.

[0073] The present invention, using alginate-chitosan nanoparticles, has been able to provide effective solutions to technical problems related to the instability of HOCl and its targeted delivery to tumor tissues. By combining the capabilities of localized and controlled HOCl production, these nanoparticles provide a significant improvement in the efficiency of anticancer treatments and can be used as a key tool in cancer treatments shortly.

[0074] Morphological study of HOCl-PFP-Alginate-Chitosan nanoparticles:

[0075] 1. Examining the morphology of nanoparticles:

[0076] Transmission electron microscope (TEM) and scanning electron microscope (SEM) techniques were used to investigate the morphology of HOCl-PFP-Alginate-Chitosan nanoparticles.

[0077] A. Transmission electron microscope (TEM):

[0078] • Sampling method: The samples were prepared by drop-casting method without using coloring. The nanoparticles were placed on a carbon-coated copper grid (400 mesh) and dried at room temperature.

[0079] • Imaging:

[0080] Imaging was performed using a transmission electron microscope (TEM, H-7650; Hitachi, Tokyo, Japan). TEM results showed that HOCl-PFP-Alginate-Chitosan nanoparticles have a core-shell structure. The core of nanoparticles is dark and the surrounding shell is light. The size of nanoparticles is relatively uniform and in the approximate range of 950 to 1100 nm.

[0081] b. Scanning electron microscope (SEM):

[0082] • Sample preparation:

[0083] HOCl-PFP-Alginate-Chitosan nanoparticles were diluted with ultra-pure water and a drop of it was placed on a Quanta 200 ESEM (FEI, USA) aluminum slide and dried at 24-40°C.

[0084] • Imaging:

[0085] Imaging was performed with a scanning electron microscope (SEM) (Hitachi S4700, Hitachi Scientific Ltd, Tokyo, Japan) at a voltage of 20 kV and a magnification of 12,000. The SEM results showed that the nanoparticles are uniformly dispersed and have a smooth surface without holes or cracks.

[0086] The answer according to DLS and TEM results:

[0087] DLS (Dynamic Light Scattering) results are consistent with SEM and TEM results, indicating that the size of HOCl-PFP-Alginate-Chitosan nanoparticles is well controlled in the expected range (950 to 1100 nm) and their morphology is by the structural design.

[0088] 2. Measurement of HOCl concentration released from nanoparticles:

[0089] A. Measurement method:

[0090] • Preparation of samples:

[0091] samples of HOCl-PFP-Alginate-Chitosan nanoparticles were prepared in different concentrations (200, 400, 800, and 1000 mg) and their volume was increased to 100 ml with autoclaved deionized water. The sample without HOCl was considered a blank sample.

[0092] • Reproducibility:

[0093] The sample was divided into four 10 ml samples and each experiment was repeated four times. The samples were subjected to sonication and their absorption was measured.

[0094] • Spectrophotometry:

[0095] HOCl concentration was measured at 234 nm and OCl- concentration at 292 nm. A quartz cuvette with a path length of 1 cm was used and the results were reported in mg/L.

[0096] b. Results:

[0097] These results include the amount of HOCl released in HOCL-PFP-Alginate-Chitosan nanoparticles for each tested concentration and can be used to evaluate the efficiency of nanoparticles in HOCl production and release.

[0098] TEM and SEM studies show proper and uniform morphology of HOCL-PFP-Alginate-Chitosan nanoparticles with core-shell structure and smooth surface. The spectrophotometric results also provide detailed information about the concentration of HOCl and OCl- released from nanoparticles, which helps to evaluate the performance and efficiency of nanoparticles in clinical applications.

[0099] Examining the release profile of HOCl using ultrasound waves:

[0100] HOCl release profile and investigation of cytotoxicity of HOCl-PFP-Alginate-Chitosan nanoparticles:

[0101] 1. HOCl release profile:

[0102] A. Test method:

[0103] • Dialysis test:

[0104] To determine the release profile of HOCl from nanoparticles, 1 ml of nanoparticles were placed in a dialysis bag with molecular weight cut-off (MWCO) of 3000 Da and 400 ml of buffer solution (pH = 6.5) at 37°C with shaking. Continuous giving was immersed in the water bath. The dialysis bag was subjected to ultrasound waves with a frequency of 1.1 MHz and an intensity of 55.8 mW/cm^2 for 10 minutes.

[0105] • Collection of samples:

[0106] At time intervals of 0, 50, 100, 150, 200, 250, 300, and 350 minutes, 1 ml of the medium was collected and replaced with 1 ml of fresh buffer.

[0107] • Results:

[0108] Release profiles showed that HOCl-PFP-Alginate-Chitosan nanoparticles release Ca(OCl)2 under the influence of ultrasound waves, which react with acid to form HOCl. Increasing the concentration of Ca(OCl)2 increases the rate of HOCl release. The results show that nanoparticles effectively release HOCl after ultrasound irradiation.

[0109] 2. Investigation of cytotoxicity:

[0110] A. Test method:

[0111] • Cell culture:

[0112] A549 cells were seeded in 96-well plates with a density of 1 × 10^4 cells per well and incubated for 12 hours.

[0113] • Treatment:

[0114] After incubation, cells were treated with PFP-Alginate-Chitosan NPs, HOCl, and HOCl-PFP-Alginate-Chitosan NPs at different concentrations for 24, 48, and 72 hours. Ultrasound waves with a power of 3 watts were applied to the cells for 15 minutes one hour after the treatment.

[0115] • MTT test:

[0116] After incubation, 20 microliters of MTT solution with a 5 mg/ml concentration were added to each well and incubated for 4 hours. After that, the culture medium was replaced with 150 μL of dimethyl sulfoxide (DMSO), and the optical density (OD) of the cells was measured at 490 nm with a Nanodrop Synergy HTX (Bio Tek, USA).

[0117] • Calculation of IC50: IC50 was calculated at 24, 48, and 72 hours using GraphPad Prism software.

[0118] b. Results:

[0119] • Cytotoxicity of nanoparticles: PFP-alginate-chitosan nanoparticles were not cytotoxic at any concentration, indicating their suitability for in-vivo applications.

[0120] • The cytotoxicity of HOCl-PFP-Alginate-Chitosan NPs: The cytotoxicity of HOCl-PFP-Alginate-Chitosan NPs increases significantly with increasing the concentration of nanoparticles and time. These nanoparticles decrease cell viability and show high lethality in A549 cells.

[0121] • Comparison with free HOCl:

[0122] Free HOCl showed higher cytotoxicity than HOCl-PFP-Alginate-Chitosan nanoparticles at the same concentration of HOCl. This suggests that the HOCl contained in the nanoparticles is gradually released, which explains the relative increase in cytotoxicity over time.

[0123] • Effects of ultrasound waves:

[0124] HOCl-PFP-Alginate-Chitosan nanoparticles showed higher cytotoxicity after using ultrasound waves (1.1 MHz; 55.8 mW/cm^2; 10 minutes) at 24, 48, and 72 hours. Also, no significant cytotoxicity of ultrasound radiation was observed in A549 cells, indicating safety. Ultrasound radiation.

[0125] c. Controls:

[0126] PBS solution and drug-free nanoparticle (PFP-Alginate-Chitosan) were used as control, to check cell viability with different concentrations of HOCl in the range of 200 to 1000 ppm.

[0127] Conclusion:

[0128] HOCl release profile shows that HOCL-PFP-Alginate-Chitosan nanoparticles release HOCl well after ultrasound irradiation. Cytotoxicity tests showed that PFP-Alginate-Chitosan nanoparticles are non-toxic, while free HOCl and HOCl-PFP-Alginate-Chitosan nanoparticles show higher cytotoxicity with increasing concentration and

time. The use of ultrasound waves also affects cytotoxicity and can be effectively used to release HOCl from nanoparticles.

[0129] Investigating the expression of proteins responsible for apoptosis and the effect of HOCl-PFP-Alginate-Chitosan NPs on the orthotopic model of lung cancer mice:

[0130] 1. Examining the expression of proteins responsible for apoptosis:

[0131] A. Test method:

[0132] • Treatment of cells: A549 cells were treated with HOCl-PFP-Alginate-Chitosan NPs for 24 and 48 hours.

[0133] • Cell lysis:

[0134] Cells were lysed using RIPA buffer at 4°C.

[0135] • Protein assay:

[0136] Protein concentration was measured using the BCA protein assay kit.

[0137] • Isolation of proteins:

[0138] Proteins were separated using SDS-PAGE.

[0139] • Western blot transfer:

[0140] After electrophoresis, the proteins were transferred to the PVDF membrane and blocked with TBST containing 5% Nonfat milk.

[0141] • Primary and secondary antibodies:

[0142] The membrane was incubated with primary antibodies including Caspase-3, PARP-1, Bax, BCL2, and GAPDH (1:1000) overnight at 4°C and then washed with TBST. After that, the membrane was incubated with peroxidase (HRP)-conjugated secondary antibody at room temperature for 1 hour. ECL reagent was used to reveal the protein band.

[0143] b. Results:

[0144] • Apoptotic proteins:

[0145] Western blot results showed that HOCl-PFP-Alginate-Chitosan NPs led to an increase in the expression of apoptotic proteins including Caspase-3, PARP-1, and Bax in A549 cells, while the expression of anti-apoptotic protein Bcl-2 decreased. These changes indicate the stimulation of apoptotic pathways by HOCl-PFP-Alginate-Chitosan NPs.

[0146] 2. Investigating the penetration and effect of HOCl-PFP-Alginate-Chitosan NPs on the orthotopic model of lung cancer mice:

[0147] A. Creation of orthotopic mouse model:

[0148] • Transfection and injection: A549 cells were transfected with luciferase and injected intrathecally into mice.

[0149] • Confirmation of deposition:

[0150] Confirmation of the initial deposition of cancer cells was performed using the LV200 LUMINOVIEW imaging system and bioluminescence technique.

[0151] b. Investigation of histology and distribution of nanoparticles:

[0152] • Tissue analysis: The lungs of mice were subjected to histological examination after being killed and dissected. The results showed that untreated mice with tumors had multiple tumor nodules compressing adjacent bronchi and alveoli, while the lungs of healthy mice had healthy epithelial tissue and open alveoli without inflammation.

[0153] • Distribution of nanoparticles:

[0154] The distribution of inhaled HOCl-PFP-Alginate-Chitosan NPs nanoparticles was investigated using a Collison nebulizer. Nanoparticles were nebulized inside the lungs with a flow rate of 2 liters per minute. The distribution and accumulation of nanoparticles in different organs were investigated up to 10 days after drug nebulization. The results showed that nanoparticles accumulate significantly in the lungs and their accumulation in other organs decreases.

[0155] Response to treatment:

[0156] • Tumor volume: a rapid increase in tumor volume was observed in both the control group and the PFP-Alginate-Chitosan NPs group. However, the administration of free HOCl or HOCl-PFP-Alginate-Chitosan NPs showed a significant inhibition in tumor growth, which effect was evident after the irradiation of ultrasound waves.

[0157] • Tumor weight: on the 10th day, the tumor weight in the groups treated with free HOCl or HOCl-PFP-Alginate-Chitosan NPs was 29.12% and 2.32% of the tumor weight of the control group.

[0158] • Survival: The survival rate of mice was measured during the experiment, and treatment with HOCl-PFP-Alginate-Chitosan nanoparticles under ultrasound radiation led to a significant increase in the survival of mice up to 40 days.

[0159] • TUNEL staining:

[0160] TUNEL staining showed that apoptosis increased in tumors of mice treated with HOCl-PFP-Alginate-Chitosan NPs and ultrasound radiation.

[0161] Conclusion:

[0162] The combination of HOCl-PFP-Alginate-Chitosan nanoparticles with ultrasound radiation effectively increased apoptosis in A549 cells and decreased tumor growth in the orthotopic model of lung cancer mice. This therapeutic combination led to preferential accumulation of nanoparticles in the lungs and decreased distribution in other organs and also showed higher therapeutic efficacy than free HOCl. The findings show that this treatment method can be considered an effective option for the treatment of lung cancer.

Advantage Effects of the Invention

[0163] 1. Targeted pulmonary delivery of HOCl:

[0164] • The new invention enables the direct and targeted delivery of hypochlorous acid (HOCl) to the lungs. This targeting specifically focuses on tumor tissues and prevents

the general distribution of the drug in other organs. This feature significantly increases the accuracy of treatment and reduces side effects.

[0165] 2. Stable production of HOCl at the site of cancer:

[0166] • This nanoproduct can produce HOCl stably at the tumor site, which means ensuring the constant and effective presence of HOCl in the tumor area. This capability effectively improves treatment efficacy compared to other methods that may release HOCl unevenly or aimlessly.

[0167] 3. The possibility of nano drug inhalation administration:

[0168] • The possibility of using the inhalation method to administer this nano drug provides direct access to the lungs and is especially suitable for local treatments in the lungs. This administration method can significantly improve drug absorption and reduce the need for high systemic doses.

[0169] 4. Significant reduction of drug side effects:

[0170] • The use of nanoparticles to release HOCl locally significantly reduces the side effects of the drug, because the drug reaches the tumor site directly and the negative effects on the surrounding healthy tissues are reduced.

[0171] 5. Being fast-acting and destroying lung cancer cells:

[0172] • Nanoparticles with controlled release of HOCl can quickly reach tumor tissues and release HOCl. This feature increases the speed of the drug's effect and quickly destroys lung cancer cells.

[0173] 6. Precise spatiotemporal control of drug release:

[0174] • This nano-drug system can precisely control the space and time for the release of HOCl. This feature helps to better manage the time and place of drug release, thereby significantly improving the effectiveness of the treatment.

[0175] 7. Controlling the drug delivery cycle:

[0176] • The nanoparticles of this invention are specifically designed to control the drug delivery cycle. This control means the possibility of planning the gradual and optimal release of HOCl during the treatment period, which increases the effectiveness and reduces the number of prescriptions.

[0177] 8. Increasing therapeutic effectiveness through pulmonary drug delivery:

[0178] • Inhaling drugs directly to the lungs and lung tumors improves the effectiveness of the treatment. This method prevents indirect drug delivery and allows more and more effective drug delivery.

[0179] 9. Releasing the drug and increasing the efficiency of penetration into the tumor through the mechanical effect caused by the radiation of ultrasound waves:

[0180] • Irradiation of ultrasound waves to nanoparticles increases drug penetration and efficiency in the tumor. This mechanical effect can help improve the distribution

and effect of HOCl within the tumor, which in turn increases the effectiveness of the treatment.

[0181] These advantages demonstrate that the claimed invention provides unique capabilities in the treatment of lung cancer over prior inventions by improving targeted delivery, reducing side effects, and increasing treatment efficacy.

Brief Description of Drawings

[0182] [Fig.1] Displays a flowchart of the experiment of HOCl-PFP-Alginate-Chitosan nanoparticles in treating lung cancer in mice.

[0183] [Fig.2] Presents a flowchart of the HOCl release profile and investigation of cytotoxicity of HOCl-PFP-Alginate-Chitosan nanoparticles and explains its testing method.

Description of Embodiments

[0184] [Fig.1] Shows a flowchart of the use of HOCl-PFP-Alginate-Chitosan nanoparticles in treating lung cancer in the orthotopic mouse model which begins by injecting transfected A549 cells into the mice, their tissue went through histology examination and confirmed their deposition using bioluminescence imaging. The distribution of nanoparticles in the lungs was analyzed, showing significant accumulation in the lungs and reduced distribution in other organs. Treatment with HOCl-PFP-Alginate-Chitosan nanoparticles and ultrasound radiation resulted in significant inhibition of tumor growth, decreased tumor weight, increased survival rate, and increased apoptosis in tumor cells. Then, the TUNEL staining showed that apoptosis increased in tumors of mice treated with HOCl-PFP-Alginate-Chitosan NPs and ultrasound radiation. The combination of nanoparticles with ultrasound radiation showed higher therapeutic efficacy than free HOCl, indicating it as a promising treatment option for lung cancer.

[0185] [Fig.2] is a flowchart of the HOCl release profile and investigation of cytotoxicity of HOCl-PFP-Alginate-Chitosan nanoparticles and explains its testing method. To determine the release profile of HOCl from nanoparticles, 1 mL of nanoparticles was placed in a dialysis bag with a molecular weight cut-off (MWCO) of 3000 Da, along with 400 mL of buffer solution (pH = 6.5) at 37°C, with shaking. The dialysis bag was continuously immersed in a water bath. Additionally, the bag was subjected to ultrasound waves with a frequency of 1.1 MHz and an intensity of 55.8 mW/cm² for 10 minutes. At time intervals of 0, 50, 100, 150, 200, 250, 300, and 350 minutes, 1 mL of the medium was collected and replaced with 1 mL of fresh buffer. Finally, the release profiles indicated that HOCl-PFP-Alginate-Chitosan nanoparticles release $Ca(OCl)_2$ under the influence of ultrasound waves, which subsequently react with acid to form HOCl. Moreover, increasing the concentration of $Ca(OCl)_2$ enhances the rate

of HOCl release. The results demonstrate that the nanoparticles effectively release HOCl following ultrasound irradiation.

Examples

[0186] By developing an advanced nano drug system for the controlled release of HOCl, this innovation aims to significantly enhance lung cancer treatment outcomes and improve the quality of life for patients with this disease.

[0187] [Formula. 1]:

[0188]

$$HOCl + {}^{\bullet}O \rightarrow {}^{\bullet}OH + O2 + Cl^{-}$$

$$HOCl + Fe2^{+} \rightarrow {}^{\bullet}OH + Fe3^{+} Cl^{-}$$

[0189] To apply the invention of the nanomedicine product with the ability to control the release of HOCl at the tumor site for the treatment of lung cancer in clinical trials, the following implementation process can be considered a suitable and efficient approach:

[0190] Implementation process for the use of nanomedicine in the treatment of lung cancer

[0191] Tumor evaluation and identification:

[0192] • First stage: advanced imaging technologies such as PET scan (positron emission tomography) are used to identify and specify the exact location and size of the tumor in the patient's lung. This step is very important because accuracy in determining the location of the tumor is critical for effective drug injection and release.

[0193] Administration of HOCl-PFP-Alginate-Chitosan nanoparticles:

[0194] • The second step: HOCl-PFP-Alginate-Chitosan NPs with a specific concentration are administered to the patient's lungs through inhalation. For this purpose, a nebulizer device is used that makes nanoparticles in the form of tiny aerosols so that they can easily enter the lungs and reach the tumor tissue.

[0195] Controlled drug release using ultrasound waves:

[0196] • The third stage: After the nanoparticles are inhaled and reach the tumor tissue, ultrasound waves with a certain frequency and intensity (for example, 1.1 MHz and 55.8 mW/cm2) are accurately and controlled to the tumor area. These ultrasound waves release HOCl from nanoparticles at the tumor site.

[0197] Care and evaluation after treatment:

[0198] • Fourth stage: The patient is under close medical supervision and regular clinical tests are performed to evaluate the response of the tumor to the treatment and the effectiveness of the drug. This includes follow-up imaging (such as a PET scan) to check for changes in tumor size and biochemical responses to treatment.

[0199] Measurement and management of side effects:

[0200] • The fifth step: possible side effects caused by the treatment are investigated and managed. This includes assessment of clinical symptoms, analysis of biological samples, and use of supportive medications to reduce adverse effects.

[0201] Summary of the process:

[0202] Tumor diagnosis: A PET scan is used to accurately identify the location of the tumor.

[0203] Administration of nanoparticles: inhalation of HOCl-PFP-Alginate-Chitosan nanoparticles into the lung.

[0204] Drug release: Using ultrasound waves to release HOCl in the tumor area.

[0205] Care: monitoring and evaluating treatment response.

[0206] Side effects management: review and management of side effects.

[0207] This operational process ensures optimization and precise coordination between advanced imaging technologies, targeted drug delivery, and controlled drug release, ultimately helping to increase treatment effectiveness and reduce side effects.

Industrial Applicability

[0208] The invention of HOCl-PFP-Alginate-Chitosan NPs has significant industrial and medical applications. This nanoparticle, with a double-layer structure that includes alginate containing PFP and chitosan conjugated with folate, and made using advanced production techniques such as Nano-emulsion and Ionic gelation, is specifically designed for the treatment of lung cancer. The industrial advantages and applications of this invention are:

[0209] Industrial and medical applications of HOCl-PFP-Alginate-Chitosan NPs:

[0210] 1. Targeted treatment of lung cancer:

[0211] • Application: This nanoparticle is specifically designed to treat lung cancer. Using the bilayer structure and its special capabilities, HOCl-PFP-Alginate-Chitosan NPs are capable of controlled release of hypochlorous acid (HOCl) at the tumor site, which improves treatment efficacy and reduces unnecessary side effects.

[0212] 2. Targeted drug delivery through inhalation:

[0213] • Application: Nanoparticles can be delivered to the lungs via inhalation, which is a delivery method that is particularly suitable for lung tumors. This process improves drug accumulation in tumor tissue and increases treatment efficiency.

[0214] 3. Increasing the effectiveness of treatment using ultrasound waves:

[0215] • Application: Using ultrasound waves to release drugs at the tumor site allows nanoparticles to penetrate the tumor tissue more effectively and release HOCl. This technology helps to increase the effectiveness of treatment and reduce the need for high doses of medicine.

[0216] 4. Application in medical centers and hospitals:

[0217] • Application: HOCl-PFP-Alginate-Chitosan NPs nanoparticles can be used as an advanced treatment option in medical centers, hospitals, and special clinics for lung cancer treatment. These nanoparticles allow doctors to provide targeted and effective treatments for patients with lung cancer.

[0218] 5. Application in cancer research centers:

[0219] • Application: These nanoparticles can also be used in cancer research centers to investigate and develop new methods of lung cancer treatment and to evaluate the effectiveness of various treatments. This possibility helps researchers to obtain more accurate scientific data about the effectiveness and safety of new treatments.

[0220] Conclusion:

[0221] The invention of HOCl-PFP-Alginate-Chitosan NPs has significant industrial and medical applications due to its special abilities in the targeted treatment of lung cancer, improving drug delivery methods, and increasing therapeutic efficacy. This nanoparticle plays an important role not only in clinical and hospital treatments but also in scientific research and is introduced as an advanced tool in the fight against lung cancer.

Claims

[Claim 1] A nanomedicine with a combination of HOCl-PFP-Alginate-Chitosan nanoparticles, which is made from a two-step method including nanoemulsion and coating for controlled delivery of HOCl to the tumor site, and increases apoptosis in A549 cells and reduces lung cancer tumor growth with ultrasound irradiation.

[Claim 2] According to claim 1, to make PFP-containing alginate nanoparticles, first 1 ml of a 5% calcium chloride solution containing 0.1% polyvinyl alcohol is combined with different concentrations of Ca(OCl)2 and 1-5% Perfluoro Pentane (PFP) solution and stirred using a sonicator probe until complete mixing is achieved.

[Claim 3] According to claim 2, after complete mixing, this compound is added to a solution of sodium alginate with a concentration of 1.23% and trehalose with a concentration of 8.25% at pH = 5 using a jet nebulizer. This process results in the formation of alginate nanoparticles containing PFP.

[Claim 4] According to claim 3, to make folate-conjugated chitosan, 44 mg of folate is dissolved in 15 ml of anhydrous DMSO, and then EDC (Dimethylaminopropyl)-N′-ethylcarbodiimide hydrochloride) is added to the solution and stirred for one hour at room temperature.

[Claim 5] According to claim 4, for conjugation, 5 mM chitosan sodium acetate 5% v/v at pH = 5 is added to the solution and stirred for 16 hours at 30°C to allow the carboxylic groups of folate to react with the amino groups of chitosan and form an amide bond.

[Claim 6] According to claim 5, the reaction mixture is dialyzed for 3 days in a sodium phosphate buffer solution with pH = 7.4 after adjusting the pH to 9.0 and then stored at -48°C for 12 hours and dried.

[Claim 7] According to claim 6, the folate-conjugated chitosan is dissolved in a PBS solution containing 2.5 mM salicylic acid.

[Claim 8] According to claim 7, 4 ml of chitosan solution containing salicylic acid is slowly added to the stirring alginate solution (at 500 rpm), and the suspension is centrifuged for 30 minutes at 20,000 rpm and then washed three times to remove impurities.

[Claim 9] According to claim 8, the folate content in HOCl-PFP-Alginate-Chitosan nanoparticles is determined using a fluorometric method and a multi-mode microplate reader.

[Claim 10] According to claim 9, 2 mg of nanoparticles are dissolved in a mixture of DMSO and dichloromethane and then the amount of folate is measured and calculated using a calibration curve, and the amount of folate conjugated to chitosan is calculated to be 78.5% ± 0.09.

[Claim 11] According to claim 10, the size distribution and zeta potential of the nanoparticles were measured using a Malvern Zeta Nano ZS device, which shows that the average hydrodynamic size of the nanoparticles was 1092 ± 170.3 nm and the relatively low particle dispersion index (PDI) (<0.3) indicates a small size distribution of the nanoparticles.

[Claim 12] According to claim 11, using high-performance liquid chromatography (HPLC), the encapsulation efficiency of the HOCl-PFP-Alginate-Chitosan-folate-NPs nanoparticles was determined to be 98.38%.

[Claim 13] According to claim 12, transmission electron microscopy (TEM) and scanning electron microscopy (SEM) techniques were also used to examine the morphology of the HOCl-PFP-Alginate-Chitosan nanoparticles.

[Claim 14] According to claim 13, for the TEM method, the samples were prepared by the Drop-Casting method without using dye, and the nanoparticles were placed on a 400 Mesh carbon-coated copper grid and dried at room temperature.

[Claim 15] According to claim 14, imaging was performed using a transmission electron microscope, and the TEM results showed that the HOCl-PFP-Alginate-Chitosan nanoparticles have a core-shell structure and the core of the nanoparticles is dark and the shell around it is light and the size of the nanoparticles is relatively uniform in the approximate range of 950 to 1100 nm.

[Claim 16] According to claim 15, for the SEM method, the HOCl-PFP-Alginate-Chitosan nanoparticles were diluted with Ultra-pure water and a drop of it was placed on the aluminum piece of Quanta 200 ESEM (FEI, USA) SEM, and dried at a temperature of 24 to 40 ° C.

[Claim 17] According to claim 16, electron microscopy imaging was performed at a voltage of 20 kV and a magnification of 12,000, and the SEM results showed that the nanoparticles were uniformly dispersed and had a smooth surface without holes or cracks.

[Claim 18] According to claim 17, the results of DLS (Dynamic Light Scattering) studies were consistent with the SEM and TEM results, and the size of the HOCl-PFP-Alginate-Chitosan nanoparticles was controlled within the expected range (950 to 1100 nm) and their morphology was predicted per the structural design.

[Claim 19] According to claim 18, to measure the concentration of HOCl released from the nanoparticles, samples of HOCl-PFP-Alginate-Chitosan nanoparticles were prepared at different concentrations (200, 400, 800, and 1000 mg) and their volume was brought to 100 ml with autoclaved deionized water, and the sample without HOCl was considered as a blank sample.

[Claim 20] According to claim 19, each sample was divided into four 10 ml samples and each experiment was repeated four times to be subjected to sonication and their absorbance was measured.

[Claim 21] According to claim 20, the HOCl concentration was measured at a wavelength of 234 nm and the OCl- concentration at 292 nm, and a quartz cuvette with a path length of 1 cm was used, and the results were reported in mg/l.

[Claim 22] According to claim 21, to determine the HOCl release profile from nanoparticles, 1 ml of nanoparticles was placed in a dialysis bag with a molecular weight cut-off (MWCO) of 3000 Da and immersed in 400 ml of buffer solution (pH = 6.5) at 37 ° C and with continuous shaking in a water bath, then the dialysis bag was subjected to ultrasound waves with a frequency of 1.1 MHz and an intensity of 55.8 mW/cm^2 for 10 minutes.

[Claim 23] According to claim 22, at time intervals of 0, 50, 100, 150, 200, 250, 300, and 350 minutes, 1 ml of the medium was collected and replaced with 1 ml of fresh buffer.

[Claim 24] According to claim 23, the release profiles showed that HOCl-PFP-Alginate-Chitosan nanoparticles released Ca(OCl)2 under the influence of ultrasound waves, which reacted with acid to form HOCl, and increasing the concentration of Ca(OCl)2 increased

the rate of HOCl release, so the nanoparticles effectively released HOCl after irradiation with ultrasound waves.

[Claim 25] According to claim 24, to investigate cytotoxicity, A549 cells were seeded in 96-well plates at a density of 10^4 × 1 cells per well and incubated for 12 hours.

[Claim 26] 26- According to claim 25, after incubation, the cells were treated with PFP-Alginate-Chitosan NPs, HOCl, and HOCl-PFP-Alginate-Chitosan NPs at different concentrations for 24, 48, and 72 hours. Ultrasound waves with a power of 3 watts were applied to the cells for 15 minutes one hour after treatment.

[Claim 27] According to claim 26, after incubation, 20 μL of MTT solution with a concentration of 5 mg/mL was added to each well and incubated for 4 hours, after which the culture medium was replaced with 150 μL of dimethyl sulfoxide (DMSO) and the optical density (OD) of the cells was measured at a wavelength of 490 nm with a nanodrop device.

[Claim 28] According to claim 27, IC50 was calculated at 24, 48, and 72 hours using GraphPad Prism software.

[Claim 29] According to claim 28, PFP-Alginate-Chitosan nanoparticles were not cytotoxic at any concentration, indicating their suitability for in-vivo applications.

[Claim 30] According to claim 29, the cytotoxicity of HOCl-PFP-Alginate-Chitosan NPs increases significantly with increasing nanoparticle concentration and time, so these nanoparticles lead to a decrease in cell viability and show high killing power in A549 cells.

[Claim 31] According to claim 30, free HOCl showed higher cytotoxicity than HOCl-PFP-Alginate-Chitosan nanoparticles at the same HOCl concentration, which means that the HOCl contained in the nanoparticles is gradually released, explaining the relative increase in cytotoxicity over time.

[Claim 32] According to claim 31, HOCl-PFP-Alginate-Chitosan nanoparticles showed higher cytotoxicity after the application of ultrasound waves (1.1 MHz; 55.8 mW/cm^2; 10 min) at 24, 48, and 72 hours, and no significant cytotoxicity from ultrasound radiation was observed in A549 cells.

[Claim 33] According to claim 32, for control, PBS solution and drug-free nanoparticles (PFP-Alginate-Chitosan) were used to examine cell

viability alongside different concentrations of HOCl in the range of 200 to 1000 ppm.

[Claim 34] According to claim 33, the expression of proteins responsible for apoptosis and the effect of HOCl-PFP-Alginate-Chitosan NPs on an orthotopic mouse model of lung cancer are examined.

[Claim 35] According to claim 34, A549 cells were treated with HOCl-PFP-Alginate-Chitosan NPs for 24 and 48 hours. And lysed using RIPA buffer at 4 ° C.

[Claim 36] According to claim 35, the concentration of proteins was measured using a BCA protein assay kit, and proteins were separated using SDS-PAGE, transferred to PVDF membrane after electrophoresis, and blocked with TBST containing 5% Nonfat milk.

[Claim 37] According to claim 36, the membrane was incubated with primary antibodies including Caspase-3, PARP-1, Bax, BCL2, and GAPDH (1:1000) for one night at 4°C and then washed with TBST, after which the membrane was incubated with peroxidase (HRP) conjugated secondary antibody at room temperature for 1 hour and ECL reagent was used to detect the protein band.

[Claim 38] According to claim 37, the results of Western blotting showed that HOCl-PFP-Alginate-Chitosan NPs increased the expression of apoptotic proteins including Caspase-3, PARP-1 and Bax in A549 cells, while the expression of the anti-apoptotic protein Bcl-2 decreased, indicating the stimulation of apoptotic pathways by HOCl-PFP-Alginate-Chitosan NPs.

[Claim 39] According to claim 38, to investigate the penetration and effect of HOCl-PFP-Alginate-Chitosan NPs on the orthotopic mouse model of lung cancer, A549 cells were transfected with luciferase and injected intrathecally into mice, and the initial deposition of cancer cells was confirmed using the LV200 LUMINOVIEW imaging system and bioluminescence technique.

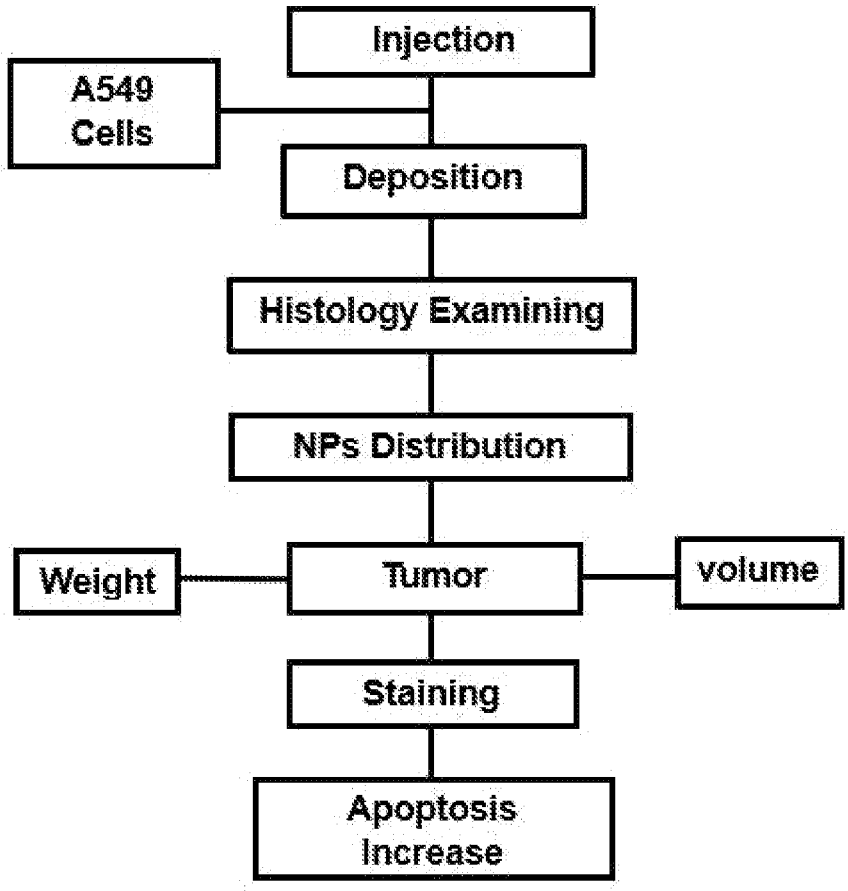
[Claim 40] According to claim 39, the lungs of the mice were subjected to histological examination after being killed and dissected, and the results showed that the untreated mice with tumors had numerous tumor nodules that compressed the adjacent bronchi and alveoli, while the lungs of the healthy mice had healthy epithelial tissue and open, non-inflammatory alveoli.

[Claim 41] According to claim 40, the distribution of inhaled HOCl-PFP-Alginate-Chitosan NPs nanoparticles was examined using a Collison nebulizer and the nanoparticles were nebulized at a flow rate of 2 liters per minute inside the lungs to examine the distribution and accumulation of nanoparticles in various organs up to 10 days after nebulization of the drug, which ultimately showed that the nanoparticles significantly accumulate in the lungs and their accumulation in other organs is reduced.

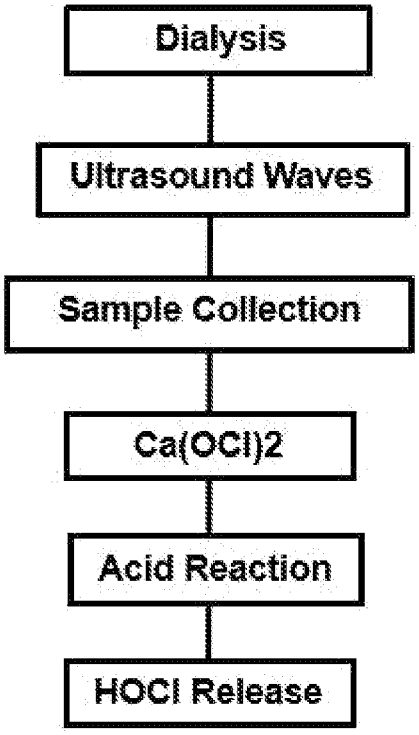
[Claim 42] According to claim 42, on day 10, the tumor weight in the groups treated with free HOCl or HOCl-PFP-Alginate-Chitosan NPs was 29.12% and 2.32% of the tumor weight in the control group, respectively.

[Claim 43] According to claim 43, the survival rate of mice was measured during the experiment, and treatment with HOCl-PFP-Alginate-Chitosan nanoparticles under ultrasound irradiation resulted in an increase in the survival of mice up to 40 days.

[Claim 44] According to claim 44, TUNEL staining showed that apoptosis was increased in tumors of mice treated with HOCl-PFP-Alginate-Chitosan NPs and ultrasound irradiation.


Injection
A549 Cells
Deposition
Histology Examining
NPs Distribution
Weight
Tumor
volume
Staining
Apoptosis Increase

[Fig. 1]


Dialysis
Ultrasound Waves
Sample Collection
Ca(OCl)2
Acid Reaction
HOCl Release

[Fig. 2]

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2024/062818

A. CLASSIFICATION OF SUBJECT MATTER
A61K47/36,A61K31/02,C08L5/04,C08L5/08,B82Y5/00,A61P35/00 Version=2025.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61K, A61P, C08L, B82Y

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

PatSeer, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	IN202311081092 A; (INDIAN INSTITUTE OF TECHNOLOGY BHU VARANASI [IN]); 13 September 2024 Abstract, claims	1-28
Y	WO2013075186 A1; (UNICAMP [BR]); 30 May 2013 Abstract, claims	1-28
Y	CN105963702 A; (UNIV HUBEI); 28 September 2016 Abstract, claims	1-28
Y	S Bhunchu et al.; Biopolymeric alginate-chitosan nanoparticles as drug delivery carriers for cancer therapy; Pharmazie 2014 Aug;69(8); 563-70. Abstract	1-28
Y	Yuhan Zhou et al.; Secretions from hypochlorous acid-treated tumor cells delivered in a melittin hydrogel potentiate cancer immunotherapy; Bioact Mater. 2021 Jul 24(9); 541-553. Abstract	1-28

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
19-05-2025	19-05-2025
Name and mailing address of the ISA/ Indian Patent Office Plot No.32, Sector 14,Dwarka,New Delhi-110075 Facsimile No.	Authorized officer Dr. Sateesh Kumar Meena Telephone No. +91-1125300200

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IB2024/062818

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☒ Claims Nos.: 29-44
because they relate to subject matter not required to be searched by this Authority, namely:
The subject matter of claims 29-44 relates to a method for the treatment of the human or animal body, which does not require an international search by the International Searching Authority in accordance with PCT Article 17(2)(a)(i) and [Rule 39.1(iv)].

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IB2024/062818

Citation	Pub.Date	Family	Pub.Date
WO 2013075186 A1	30-05-2013	BR PI1105020 A2	22-04-2014